

Rise of AI in development

A primer on how AI will tie into the developer workflow

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We had the privilege of talking to some of the key executives at GitHub during GitHub Universe 2023 where they announced some of the biggest platform changes since the day GitHub was created, and AI plays a huge part. Git was so revolutionary to the every software developer's journey that software version control is something every developer is familiar with and none can match the popularity of GitHub. With most of the developers across the world using GitHub, the insights gleaned from GitHub form a very accurate representation of the latest happenings in the world of software development. By 2023, GitHub's data revealed a significant shift towards AI integration in development processes, echoing Git's foundational impact but in the realm of AI.

To not be left behind, even GitHub has incorporated several AI-based features which look to completely disrupt how software is written. Announcing the launch of these features at GitHub Universe 2023, their annual conference, Thomas Dohmke, CEO - GitHub stated, "Just as GitHub was founded on Git, today we are re-founded on Copilot."

AI in development

AI's integration into GitHub represents more than a technological trend; it signifies a paradigm shift in software development. "In the last year, we noticed that a lot of data manipulation languages shot up through our Octaverse report. So

languages such as TeX and T-SQL which are generally used for data manipulation, saw an uptick which makes sense when you're building more and more AI," said Kyle Daigle, COO - GitHub.

This adoption is indicative of a deeper, more transformative integration into the fabric of software development. The embrace of generative AI, evidenced by a surge in open-source AI projects—some even ranking among the top 10 by contributor count—highlights a growing experimentation and innovation ethos within the developer community.

Generative AI, particularly through foundation models from OpenAI and other entities, has also propelled open-source projects to unprecedented popularity. This burgeoning interest is a testament to the developer community's role as a catalyst for AI innovation, with GitHub serving as a pivotal platform for exploration, adoption, and dissemination of AI technologies.

Mario Rodriguez, VP of Product, GitHub explained how AI made its way into every stage of the developer journey on GitHub and how it's going to evolve, "We want AI platform to be on four foundational pillars. One of them is AI and compiler need to be ubiquitous across the entirety of the lifecycle. That means that it is where you expect it to be, and it should be doing these things for you. It should be in the AI... It should be in the PR flow... It should be in the debugger flow... It should be in all of these places. That's why you see us in execution right now. It started in the IDE and was present only in code completion. But now it's in chat. Now it's available on your phone. Now it's available on your CLI. Now it's available on github.com. Now it's available on a diff as well, in a PR. So it's like in a bunch of these places already. The other thing is about customisation, conversation and natural language. These models are just amazing at natural language processing. I'm really taking something semantic, understanding that, and then give you an answer back. So we're going to make a big bet on that. And the third thing is personalisation. That's what we talked about with our embeddings, with fine-tuning, and things like that. And then the last thing for us is about safety and trust, because we want AI to do good for humanity. So we have to continue to invest heavily on that."

At the top of the biggest announcements at Universe 2023 was GitHub Copilot Chat. It's powered by GPT-4 so you're getting all the rich natural

